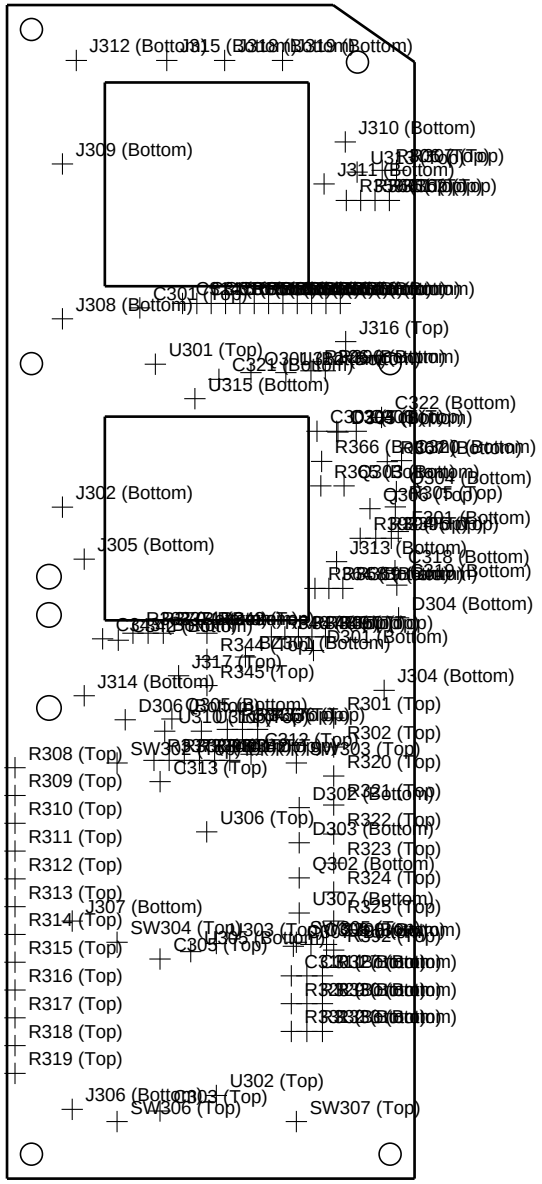


Board extent 50.000 mm (X) x 144.000 mm (Y)

Top view, Edge.Cuts and holes of CTRL.kicad_pcb; crosses = CPL positions (KiCad footprint origins). Not to a fixed scale.
NOT RELEASED: model/CI only; bench gates OPEN

## CTRL fab notes

**NOT RELEASED: model/CI only; bench gates OPEN.**
Board status: DESIGN_NOT_FOR_FABRICATION. Every value below is a CI/model output of pcb/fab_export.py and pcb/board_gen.py; no hardware or bench gate is closed by it.

## Board

Outline (Edge.Cuts of CTRL.kicad_pcb): 50.000 x 144.000 mm, area 5915.0 mm2; lower-left corner at X 100.000, Y -244.000 mm in the Gerber/CPL frame (KiCad absolute origin, Y up).
Holes: NPTH 2.70 mm, NPTH 3.00 mm (9 mounting holes; NPTH unplated and without net).

## Fabrication

Copper layers: 4; finished thickness 1.6 mm (JLC tolerance +/-10 % for boards >= 1.0 mm).
Stack-up JLC04161H-7628 (https://jlcpcb.com/impedance, read 2026-11-07; pcb/generated/netclasses.json stackup (pcb/stackup.py)):
L1 F.Cu: copper 0.035 mm
prepreg 7628x1: 0.2104 mm, Er 4.4
L2 In1.Cu: copper 0.0152 mm
core: 1.065 mm, Er 4.6
L3 In2.Cu: copper 0.0152 mm
prepreg 7628x1: 0.2104 mm, Er 4.4
L4 B.Cu: copper 0.035 mm
Layer thicknesses sum to 1.5862 mm (computed). Copper weights: outer 0.035 mm = 1 oz, inner 0.0152 mm = 0.5 oz finished (JLC "H/HOZ").
Impedance: no RF50 net on this board; order without impedance control.
Surface finish, solder mask and legend colours: order options, not fixed by this fab set.
KiCad 10.0.7 DRC on the committed board: 0 errors, 0 unconnected, 307 reviewed warnings, ignored checks [] (report.json; CI/model result).

## Files (fab/)

gerber/: Gerber X2 with netlist attributes, Protel extensions, one file per layer, Gerber job file (.gbrjob). drill/: Excellon PTH and NPTH files (mm, decimal, absolute origin, alternate oval mode) and Gerber X2 drill maps.
CTRL.ipc356: IPC-D-356 netlist of every net with a pad. assembly/: JLC BOM and CPL, the raw KiCad position file, the rotation offset table and the instance map.

## Assembly

Boards to build: 1 = instances (CTRL 1) + spares 0; every board is the same design and is assembled the same.
Sides with parts: Bottom (SMD 57, THT 17), Top (SMD 70, THT 2).
CPL: Mid X / Mid Y are the KiCad footprint origins (JLC defines them as the component centroid) and Rotation the raw KiCad rotation (degrees, counter-

clockwise as the JLC CPL column). JLC rotation and centroid offsets per
footprint are TBD (assembly/CTRL-cpl-rotation-offsets.csv): check every part
in the JLC viewer at order time; nothing here verifies them.
BOM TBD: U305 (Raspberry Pi Pico SC0915): TBD: no build/10-jlc-bom.csv row (design/parts/parts.json tbd or not a PCB mount)
BOM TBD: BZ301 (CPT-1207-3TH): TBD: build/10-jlc-bom.csv row has no LCSC code

## Not proved here (OPEN)

- that the schematic is right
- current capacity or temperature rise beyond the chosen class widths
- impedance, RF loss or isolation, EMC or signal integrity
- creepage beyond the encoded clearances
- that the enclosure fits (only outline and hole checks; heights are not checked here)
- that the fab's DFM accepts the board, or that CPL rotations are right
- that the land patterns match the purchased MPNs
- any function: hardware and bench gates stay OPEN

Sources (read 2026-11-07): https://jlcpcb.com/help/article/bill-of-materials-for-pcb-assembly; https://jlcpcb.com/help/article/pick-place-file-for-pcb-assembly; https://jlcpcb.com/help/article/how-to-generate-the-bom-and-centroid-file-from-kicad; https://jlcpcb.com/help/article/how-to-generate-gerber-and-drill-files-in-kicad-8; https://jlcpcb.com/capabilities/pcb-capabilities